

Qns	Answer	Marks																
3 (a)(i)	$(W =) 2.60 \times 10^5 \times (3.80 - 2.30) \times 10^{-3} = 390 \text{ J}$	A1																
3(a)(ii)	no (total) change (in internal energy) gas returns to its original temperature/ state/ no change to pressure and volume	B1 B1																
3(b)	A to B row all correct (1370, – 390, 980) B to C row all correct (0, 550, 550) C to A row: ΔU adds to the other two ΔU values to give zero C to A row: $w = 0$ and q adds to w to give ΔU value complete correct answer: <table border="1"><thead><tr><th>change</th><th>q / J</th><th>w / J</th><th>$\Delta U / \text{J}$</th></tr></thead><tbody><tr><td>A to B</td><td>(+)1370</td><td>–390</td><td>(+)980</td></tr><tr><td>B to C</td><td>0</td><td>(+)550</td><td>(+)550</td></tr><tr><td>C to A</td><td>–1530</td><td>0</td><td>–1530</td></tr></tbody></table>	change	q / J	w / J	$\Delta U / \text{J}$	A to B	(+)1370	–390	(+)980	B to C	0	(+)550	(+)550	C to A	–1530	0	–1530	B1 B1 B1 B1
change	q / J	w / J	$\Delta U / \text{J}$															
A to B	(+)1370	–390	(+)980															
B to C	0	(+)550	(+)550															
C to A	–1530	0	–1530															
3(c)	when vapourising, greater change in separation of atoms or molecules/ bonds are completely broken (compared to partially broken), hence a greater change in potential energy and internal energy greater change in volume, hence a greater work done	B1 B1																

